

Box 9: Pairs of Dice — Application Example

Application Example: Pairs of Dice

Consider two pairs of coins:

- **First pair (fair):**

$$\theta_1(\text{H}) = \theta_1(\text{T}) = \frac{1}{2}.$$

- **Second pair (biased):**

$$\theta_2(\text{H}) = \frac{2}{3}, \quad \theta_2(\text{T}) = \frac{1}{3}.$$

One of the two pairs is chosen at random with probability 0.5 and thrown $N = 10$ times. The sum of the two coins is recorded for each throw.

1. If the throw results in the sequence “1010101010”, which pair of dice was picked for the throw?
2. Repeat (1) for the sequence “2222200000”.
3. Assume that the result of the second coin is flipped (F) with probability $p = 0.5$ before recording the result of the throw. Repeat (1) and (2) under this assumption. Consider also the case for which $p = 0.2$.

The graphical model associated with this problem is shown in Figure 1.

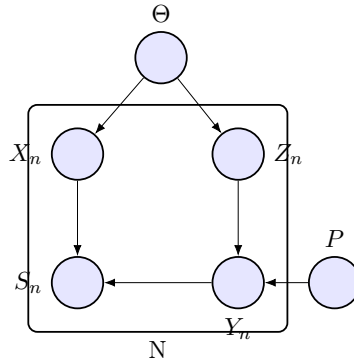


Figure 1: Graphical model for the pairs of dice problem.

- X_n represents the n -th result from the throw of the first coin, with $X_n(\text{H}) = 1$ and $X_n(\text{T}) = 0$, and

$$p(x_n | \theta) = \theta^{x_n} (1 - \theta)^{(1-x_n)}.$$

- Z_n represents the *unobserved* result from the n -th throw of the second coin, with $Z_n(\text{H}) = 1$ and $Z_n(\text{T}) = 0$, and

$$p(z_n | \theta) = \theta^{z_n} (1 - \theta)^{(1-z_n)}.$$

- Y_n represents the *recorded* result from the n -th throw of the second coin, with $Y_n(\mathbf{H}) = 1$ and $Y_n(\mathbf{T}) = 0$, and

$$p(y_n | z_n, p) = p^{I(y_n \neq z_n)}(1 - p)^{I(y_n = z_n)}.$$

- S_n is a deterministic function of X_n and Y_n :

$$S_n = X_n + Y_n, \quad p(s_n | x_n, y_n) = \mathbf{I}(s_n = x_n + y_n),$$

where

$$I(e) = \begin{cases} 1, & \text{if } e = \text{true}; \\ 0, & \text{otherwise.} \end{cases}$$

Joint Distribution

Let $\mathbf{X} = (X_1, \dots, X_N)$, $\mathbf{Y} = (Y_1, \dots, Y_N)$, $\mathbf{Z} = (Z_1, \dots, Z_N)$, $\mathbf{S} = (S_1, \dots, S_N)$, and define

$$f_n = I(y_n \neq z_n) = 1 - I(y_n = z_n), \quad \mathbf{f} = (f_1, \dots, f_N).$$

Here f_n indicates whether the second coin was flipped.

The joint probability distribution for $\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}$ given θ and p is

$$\begin{aligned} p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}, \theta, p) &= p(\theta) p(p) \prod_{n=1}^N p(x_n | \theta) p(z_n | \theta) p(y_n | z_n, p) p(s_n | x_n, y_n), \\ p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} | \theta, p) &= \frac{p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s}, \theta, p)}{p(\theta) p(p)} \\ &= \prod_{n=1}^N p(x_n | \theta) p(z_n | \theta) p(y_n | z_n, p) p(s_n | x_n, y_n) \\ &= \prod_{n=1}^N \theta^{x_n} (1 - \theta)^{(1-x_n)} \theta^{z_n} (1 - \theta)^{(1-z_n)} p^{I(y_n \neq z_n)} (1 - p)^{I(y_n = z_n)} I(s_n = x_n + y_n) \\ &= \theta^{\sum_{n=1}^N x_n} (1 - \theta)^{\sum_{n=1}^N (1-x_n)} \theta^{\sum_{n=1}^N z_n} (1 - \theta)^{\sum_{n=1}^N (1-z_n)} p^{\sum_{n=1}^N I(y_n \neq z_n)} (1 - p)^{\sum_{n=1}^N I(y_n = z_n)}. \end{aligned}$$

Sufficient Statistics

Notice that

$$x_n = s_n - y_n, \quad y_n = z_n(1 - f_n) + (1 - z_n)f_n,$$

and therefore

$$x_n = s_n - z_n + 2z_nf_n - f_n.$$

Define the sufficient statistics:

$$\begin{aligned} N_x &= \sum_{n=1}^N (s_n - z_n + 2z_nf_n - f_n) = N_s - N_z + 2N_{zf} - N_f \geq 0, \\ N_s &= \sum_{n=1}^N s_n, \quad N_z = \sum_{n=1}^N z_n, \\ N_{zf} &= \sum_{n=1}^N z_nf_n, \quad N_f = \sum_{n=1}^N f_n. \end{aligned}$$

Therefore, the probability distribution function can be written as

$$\begin{aligned} p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) &= \theta^{N_x} (1 - \theta)^{(N - N_x)} \theta^{N_z} (1 - \theta)^{(N - N_z)} p^{N_f} (1 - p)^{N - N_f} \\ &= \theta^{N_x + N_z} (1 - \theta)^{2N - (N_x + N_z)} p^{N_f} (1 - p)^{N - N_f} \\ &= p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p). \end{aligned}$$

Observe that $N_x = N_x(\mathbf{s}, \mathbf{z}, \mathbf{f})$ and $N_z = N_z(\mathbf{z})$. Hence,

$$p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) = p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p).$$

Also,

$$p(\mathbf{s} \mid \theta, p) = \sum_{\mathbf{x}} \sum_{\mathbf{z}} \sum_{\mathbf{y}} p(\mathbf{x}, \mathbf{z}, \mathbf{y}, \mathbf{s} \mid \theta, p) = \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p).$$

Posterior and Decision Rule

$$\begin{aligned} p(\theta \mid \mathbf{s}, p) &= \frac{p(\mathbf{s}, \theta, p)}{p(\mathbf{s}, p)} = \frac{p(\mathbf{s} \mid \theta, p) p(\theta, p)}{p(\mathbf{s}, p)} \\ &= \frac{p(\mathbf{s} \mid \theta, p) p(\theta \mid p) p(p)}{p(\mathbf{s} \mid p) p(p)} = \frac{p(\mathbf{s} \mid \theta, p) p(\theta \mid p)}{p(\mathbf{s} \mid p)} \\ &= \frac{p(\mathbf{s} \mid \theta, p) p(\theta)}{p(\mathbf{s} \mid p)}, \end{aligned}$$

since $\theta \perp\!\!\!\perp p$.

Decision Rule

Choose θ_1 if

$$\mathbb{I}\left(\frac{p(\theta_1 \mid \mathbf{s}, p)}{p(\theta_2 \mid \mathbf{s}, p)} > 1\right) \Rightarrow \text{choose } \theta_1,$$

which is equivalent to

$$\mathbb{I}\left(\frac{p(\mathbf{s} \mid \theta_1, p) p(\theta_1)}{p(\mathbf{s} \mid \theta_2, p) p(\theta_2)} > 1\right) \Rightarrow \text{choose } \theta_1,$$

since

$$p(\theta \mid \mathbf{s}, p) \propto p(\mathbf{s} \mid \theta, p) p(\theta) = \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p) p(\theta).$$

Notice that

$$p(\mathbf{s} \mid p) = \sum_{\theta} \sum_{\mathbf{f}} \sum_{\mathbf{z}} p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p) p(\theta), \quad \theta \in \{\theta_1, \theta_2\}.$$

Results for Question 3

These equations lead to the following results for question 3:

$$\begin{aligned} \frac{p([2, 2, 2, 2, 2, 0, 0, 0, 0, 0] \mid \frac{2}{3}, \frac{1}{2}) p(\frac{2}{3})}{p([2, 2, 2, 2, 2, 0, 0, 0, 0, 0] \mid \frac{1}{2}, \frac{1}{2}) p(\frac{1}{2})} &= 1.80 \Rightarrow \theta_2 = \frac{2}{3}, \\ \frac{p([1, 0, 1, 0, 1, 0, 1, 0, 1, 0] \mid \frac{2}{3}, \frac{1}{2}) p(\frac{2}{3})}{p([1, 0, 1, 0, 1, 0, 1, 0, 1, 0] \mid \frac{1}{2}, \frac{1}{2}) p(\frac{1}{2})} &= 0.13 \Rightarrow \theta_1 = \frac{1}{2}. \end{aligned}$$

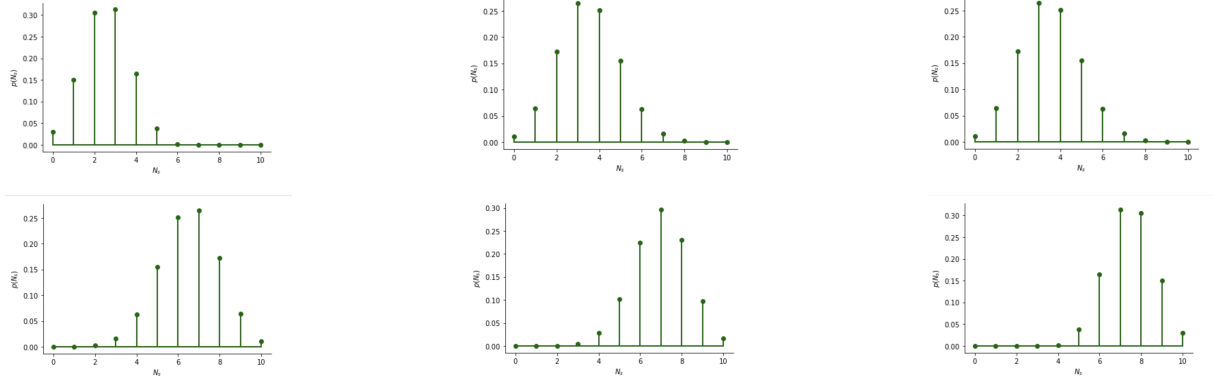


Figure 2: From left to right and from top to bottom, probability distributions $p(N_s)$ for $\theta = \frac{1}{100}, \frac{1}{9}, \frac{1}{5}, \frac{99}{100}, \frac{8}{9}, \frac{4}{5}$ for problem 3 in Box 9. The corresponding mean values, μ_{N_s} , are 2.55, 3.06, 3.5, 6.5, 6.9, and 7.45, respectively. The corresponding entropies, $E(N_s)$, are 2.44, 2.56, 2.56, 2.44, and 2.22, respectively.

Effect of θ on N_s

Also, notice that

$$\begin{aligned} E[\mathbf{s} \mid \theta, p] &= \sum_{\mathbf{s}} \sum_{\mathbf{f}} \sum_{\mathbf{z}} \mathbf{s} p(\mathbf{s}, \mathbf{z}, \mathbf{f} \mid \theta, p), \\ E[N_s \mid \theta, p] &= \sum_{N_s} N_s p(N_s \mid \theta, p). \end{aligned}$$

The *effect* of θ on N_s is

$$e(\theta_1, \theta_2 \mid p) = E[N_s \mid \theta_1, p] - E[N_s \mid \theta_2, p].$$

Distribution of N_s

Definition

$N_s = \sum_{n=1}^N S_n$ is the **total sum of the recorded values across all N throws**, where each $S_n = X_n + Y_n \in \{0, 1, 2\}$ is the sum of the two coins on throw n .

So N_s is the sum of N i.i.d. random variables taking values in $\{0, 1, 2\}$, and therefore:

$$N_s \in \{0, 1, 2, \dots, 2N\},$$

with the exact shape depending on θ (the coin bias) and p (the flip probability).

Marginal distribution of a single throw S_n

For a **single throw** $S_n = X_n + Y_n$, since $X_n, Z_n \sim \text{Bernoulli}(\theta)$ i.i.d., and $Y_n = Z_n \oplus F_n$ where $F_n \sim \text{Bernoulli}(p)$:

$$P(S_n = 0) = P(X_n = 0) \cdot P(Y_n = 0) = (1 - \theta)[(1 - \theta)(1 - p) + \theta p],$$

$$P(S_n = 1) = (1 - \theta)[\theta(1 - p) + (1 - \theta)p] + \theta[(1 - \theta)(1 - p) + \theta p],$$

$$P(S_n = 2) = \theta \cdot [\theta(1 - p) + (1 - \theta)p].$$

For the special case $p = 0$ (no flipping), these simplify to

$$P(S_n = 0) = (1 - \theta)^2, \quad P(S_n = 1) = 2\theta(1 - \theta), \quad P(S_n = 2) = \theta^2,$$

i.e., $S_n \sim \text{Binomial}(2, \theta)$ when $p = 0$ — the sum of two i.i.d. Bernoulli(θ) variables.

For $p = 1/2$, the flip makes Y_n uniform, so

$$P(S_n = 0) = \frac{1 - \theta}{2}, \quad P(S_n = 1) = \frac{1}{2}, \quad P(S_n = 2) = \frac{\theta}{2}.$$

The distribution of N_s

Sum of i.i.d. variables. Since the N throws are independent and identically distributed,

$$N_s = \sum_{n=1}^N S_n, \quad S_n \stackrel{\text{i.i.d.}}{\sim} \text{pmf}(s; \theta, p),$$

so N_s is a **sum of N i.i.d. discrete random variables on $\{0, 1, 2\}$** .

Exact pmf (via convolution). The exact pmf is the N -fold convolution of the single-throw pmf:

$$p(N_s = k \mid \theta, p) = \sum_{\substack{a, b \geq 0 \\ a + b \leq N \\ a + 2b = k}} \binom{N}{a, b, N - a - b} p_0^{N-a-b} p_1^a p_2^b,$$

where $p_0 = P(S_n = 0)$, $p_1 = P(S_n = 1)$, $p_2 = P(S_n = 2)$ as above, a is the number of throws with $S_n = 1$, b is the number of throws with $S_n = 2$, $c = N - a - b$ is the number of throws with $S_n = 0$, and

$$\binom{N}{a, b, c} = \frac{N!}{a! b! c!}$$

is the multinomial coefficient. This matches exactly the `probability_distribution_of_Ns` computation in the code.

Mean and variance. Using linearity of expectation, and because

$$Y_n \sim \text{Bernoulli}(\theta(1 - p) + (1 - \theta)p),$$

we have

$$\mathbb{E}[S_n] = \mathbb{E}[X_n] + \mathbb{E}[Y_n] = \theta + \theta(1 - p) + (1 - \theta)p = 2\theta(1 - p) + p.$$

Therefore,

$$\boxed{\mathbb{E}[N_s] = N[2\theta(1 - p) + p].}$$

For $p = 0$, this reduces to the familiar $\mathbb{E}[N_s] = 2N\theta$.

The variance of a single throw (using independence of X_n and Y_n) is

$$\text{Var}(S_n) = \text{Var}(X_n) + \text{Var}(Y_n) = \theta(1 - \theta) + [\theta(1 - p) + (1 - \theta)p][1 - \theta(1 - p) - (1 - \theta)p],$$

and because the throws are i.i.d.,

$$\boxed{\text{Var}(N_s) = N \cdot \text{Var}(S_n).}$$

For $p = 0$, this reduces to $\text{Var}(N_s) = 2N\theta(1 - \theta)$ — the binomial variance.

Entropy. The entropy of N_s (as reported in the marginal figure caption) is

$$E(N_s) = - \sum_{k=0}^{2N} p(N_s = k) \log p(N_s = k).$$

The maximum possible entropy is $\log(2N + 1)$ (uniform on $2N + 1$ values). For $N = 10$, that is $\log 21 \approx 3.04$. The reported values (2.22 to 2.56) are below this, indicating the distribution is concentrated.

Behavior in the limits.

- $p = 0$ and $\theta = 1/2$: $S_n \sim \text{Binomial}(2, 1/2)$, so $N_s \sim \text{Binomial}(20, 1/2)$. Mean = 10, variance = 5, entropy ≈ 2.71 .
- $p = 1/2$: Y_n is uniform, so S_n has pmf $((1 - \theta)/2, 1/2, \theta/2)$. The distribution becomes more spread out (closer to uniform) and the entropy increases.
- $\theta \rightarrow 0$ or $\theta \rightarrow 1$: The distribution concentrates near $N_s = 0$ or $N_s = 2N$ (or near $N_s = N$ when p dominates).
- $N \rightarrow \infty$: By the Central Limit Theorem (since $\text{Var}(S_n) > 0$),

$$\frac{N_s - N\mu_S}{\sqrt{N} \sigma_S} \xrightarrow{d} \mathcal{N}(0, 1),$$

so N_s becomes approximately normal with the mean and variance given above.

Intuition

The distribution of N_s is essentially a **compromise between two binomials**:

- The first coin contributes $X_n \sim \text{Bernoulli}(\theta)$, giving a $\text{Binomial}(N, \theta)$ contribution.
- The second coin contributes Y_n , which is a **flipped** version of $Z_n \sim \text{Bernoulli}(\theta)$. The flip with probability p effectively **mixes** Z_n toward $\text{Bernoulli}(1/2)$ as $p \rightarrow 1/2$.

When $p = 0$: $Y_n = Z_n \sim \text{Bernoulli}(\theta)$, so $N_s \sim \text{Binomial}(2N, \theta)$.

When $p = 1/2$: $Y_n \sim \text{Bernoulli}(1/2)$ independent of θ , so N_s is the sum of $\text{Binomial}(N, \theta)$ and $\text{Binomial}(N, 1/2)$ — a Poisson-binomial type distribution with two different success probabilities.

When $p = 1$: $Y_n = 1 - Z_n \sim \text{Bernoulli}(1 - \theta)$, so $N_s \sim \text{Binomial}(N, \theta) + \text{Binomial}(N, 1 - \theta)$.

This is why the “effect of θ on N_s ” matters for classification: **as p grows, the second coin carries less information about θ** , and the two hypotheses $\theta_1 = 1/2$ vs. $\theta_2 = 2/3$ become harder to distinguish. This is visible in the posterior ratios: at $p = 0$, the sequence 2222200000 gives a ratio of 1.80 in favor of θ_2 ; at larger p , the ratio shrinks toward 1.

Detailed derivation of the convolution

Step 1: Single-throw pmf. Let

$$p_k := P(S_n = k \mid \theta, p), \quad k \in \{0, 1, 2\}.$$

Explicitly, from the previous subsection,

$$\begin{aligned} p_0 &= (1 - \theta)[(1 - \theta)(1 - p) + \theta p], \\ p_1 &= (1 - \theta)[\theta(1 - p) + (1 - \theta)p] + \theta[(1 - \theta)(1 - p) + \theta p], \\ p_2 &= \theta[\theta(1 - p) + (1 - \theta)p]. \end{aligned}$$

We have $p_0 + p_1 + p_2 = 1$.

Step 2: Independence across throws. The N throws are i.i.d., so the joint pmf of $(S_1, \dots, S_N)$ is

$$P(S_1 = s_1, \dots, S_N = s_N \mid \theta, p) = \prod_{n=1}^N p_{s_n}, \quad s_n \in \{0, 1, 2\}.$$

Step 3: Summation. The event $\{N_s = k\}$ is the disjoint union of all outcomes $(s_1, \dots, s_N)$ with $\sum_{n=1}^N s_n = k$. Therefore,

$$p(N_s = k \mid \theta, p) = \sum_{\substack{s_1, \dots, s_N \in \{0, 1, 2\} \\ s_1 + \dots + s_N = k}} \prod_{n=1}^N p_{s_n}.$$

Step 4: Grouping by counts. Any configuration $(s_1, \dots, s_N)$ with sum k is uniquely characterized by the triple

$$a := \#\{n : s_n = 1\}, \quad b := \#\{n : s_n = 2\}, \quad c := \#\{n : s_n = 0\},$$

subject to the constraints

$$a + b + c = N, \quad a + 2b = k, \quad a, b, c \geq 0.$$

The number of sequences $(s_1, \dots, s_N) \in \{0, 1, 2\}^N$ realizing a given triple (a, b, c) is the multinomial coefficient

$$\binom{N}{a, b, c} = \frac{N!}{a! b! c!}.$$

The probability of each such sequence is

$$p_1^a p_2^b p_0^c.$$

Hence, grouping the sum in Step 3 by (a, b, c) :

$$p(N_s = k \mid \theta, p) = \sum_{\substack{a, b, c \geq 0 \\ a + b + c = N \\ a + 2b = k}} \frac{N!}{a! b! c!} p_1^a p_2^b p_0^c.$$

The constraints $a + b + c = N$ and $a + 2b = k$ allow us to eliminate $c = N - a - b$, giving the equivalent form

$$p(N_s = k \mid \theta, p) = \sum_{\substack{a, b \geq 0 \\ a + b \leq N \\ a + 2b = k}} \binom{N}{a, b, N - a - b} p_0^{N - a - b} p_1^a p_2^b.$$

Step 5: Interpretation as an N -fold convolution. Let $\mu = p_0\delta_0 + p_1\delta_1 + p_2\delta_2$ be the single-throw probability measure on $\{0, 1, 2\}$. Then the pmf of $N_s = \sum_{n=1}^N S_n$ is the N -fold convolution

$$\mu^{*N} = \underbrace{\mu * \mu * \dots * \mu}_{N \text{ times}},$$

where $(\mu * \nu)(k) = \sum_j \mu(j) \nu(k - j)$. Equivalently, in generating-function form, if

$$G(z) := p_0 + p_1 z + p_2 z^2$$

is the probability generating function of a single throw, then the pgf of N_s is

$$G_{N_s}(z) = [G(z)]^N = (p_0 + p_1 z + p_2 z^2)^N.$$

The coefficient of z^k in this polynomial is exactly $p(N_s = k \mid \theta, p)$.

Step 6: Generalization to unequal single-throw pmfs. If the single-throw pmfs are allowed to depend on n (i.e., S_n has pmf $p_k^{(n)}$ for $k \in \{0, 1, 2\}$), the same argument gives the *Poisson-binomial* generalization:

$$p(N_s = k) = \sum_{\substack{a, b \geq 0 \\ a+b \leq N \\ a+2b=k}} \sum_{\substack{\mathcal{A}, \mathcal{B} \text{ disjoint} \\ |\mathcal{A}|=a, |\mathcal{B}|=b}} \prod_{n \in \mathcal{A}} p_1^{(n)} \prod_{n \in \mathcal{B}} p_2^{(n)} \prod_{n \notin \mathcal{A} \cup \mathcal{B}} p_0^{(n)}.$$

For the i.i.d. case studied here, all $p_k^{(n)} = p_k$ and the inner sum collapses to the multinomial coefficient $\binom{N}{a, b, N-a-b}$.

Step 7: Connection with the code. In `probability_distribution_of_Ns`, the variable `count` computes

$$\text{count} = \#\{(s_1, \dots, s_N) \in \{0, 1, 2\}^N : s_1 + \dots + s_N = N_s\} = \sum_{\substack{a, b \geq 0 \\ a+b \leq N \\ a+2b=N_s}} \binom{N}{a, b, N-a-b},$$

by iterating over a (the number of ones) and b (the number of twos) and, for each valid pair, adding $\binom{N}{a} \binom{N-a}{b}$. Since

$$\binom{N}{a} \binom{N-a}{b} = \frac{N!}{a! (N-a)!} \cdot \frac{(N-a)!}{b! (N-a-b)!} = \frac{N!}{a! b! (N-a-b)!} = \binom{N}{a, b, N-a-b},$$

the code's count matches the multinomial coefficient exactly. The likelihood term `marginal_likelihood_fast(s)` depends only on the sum $N_s = \sum_n s_n$, so multiplying `count` by this likelihood gives the aggregated probability

$$p(N_s = k \mid \theta, p) = \text{count}(k) \cdot p(\mathbf{s} \mid \theta, p) \quad \text{for any } \mathbf{s} \text{ with } \sum_n s_n = k.$$

Step 8: Sanity check. As a consistency check, sum the pmf over k :

$$\sum_{k=0}^{2N} p(N_s = k \mid \theta, p) = \sum_{k=0}^{2N} \sum_{\substack{a, b \geq 0 \\ a+b \leq N \\ a+2b=k}} \binom{N}{a, b, N-a-b} p_0^{N-a-b} p_1^a p_2^b.$$

Interchanging the order of summation and using the multinomial theorem,

$$\sum_{\substack{a, b, c \geq 0 \\ a+b+c=N}} \binom{N}{a, b, c} p_0^c p_1^a p_2^b = (p_0 + p_1 + p_2)^N = 1,$$

confirming that the pmf is normalized.